Conformal Prediction for Reliable Exoplanet Candidate Classification Milestone Report

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1 Introduction

Exoplanet detection from stellar light curves is a classification problem where false positives can be very costly, because promising candidates may require additional analysis or follow-up observation. Traditional neural-network classifiers can achieve strong predictive accuracy, but they may still produce confident predictions on ambiguous or noisy inputs. This creates a reliability problem. In practical candidate flagging, it is often better for a model to flag an uncertain case for further review than to confidently classify a false positive as a confirmed planet.

Our project investigates whether conformal prediction can improve the reliability of a CNN that classifies Kepler light curves as confirmed planets or false positives. Since the original proposal, the scope has been narrowed to focus first on Kepler KOI data and on measuring the tradeoff between false-positive reduction and model abstention before expanding to broader distribution-shift conditions. The current thesis of the project is that conformal prediction can reduce confident false-positive decisions by allowing the model to abstain on uncertain light curves while maintaining useful prediction coverage.

2 Data and Preprocessing

We constructed the dataset using the NASA Kepler Objects of Interest (KOI) catalog. From the catalog, we sampled 100 confirmed exoplanets and 100 false positives to create a balanced binary classification dataset. The corresponding Kepler light curves were downloaded using each target's Kepler ID. Of the 200 selected targets, 197 were successfully processed into model-ready examples.

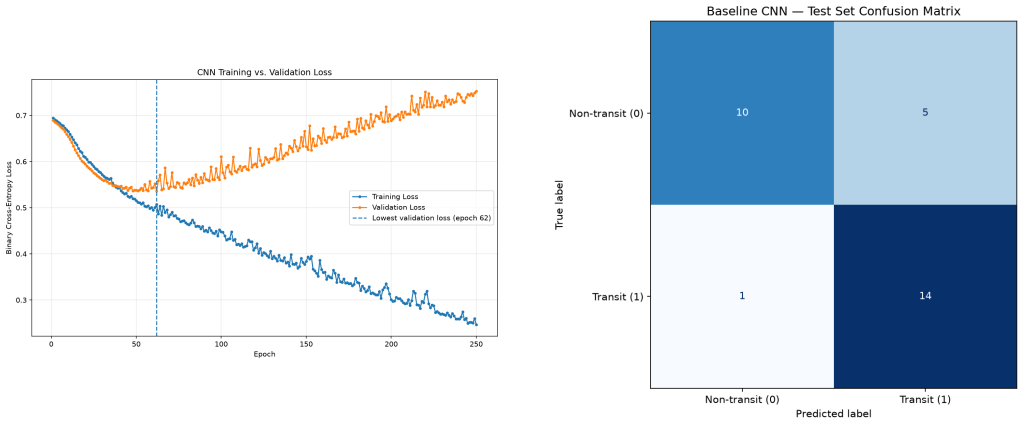
Each light curve was cleaned by removing missing values, flattening long-term trends, and filtering large positive outliers. The observations for each target were then combined and folded using the KOI catalog period and transit epoch so that repeated transit events were aligned. To create a consistent input size for the neural network, each folded light curve was binned and interpolated into a fixed-length representation of 500 points. The resulting flux values were centered and standardized for model training.

The final dataset was divided into separate training, validation, calibration, and test sets. Approximately 60% of the data was used for training, 15% for validation, 10% for conformal calibration, and 15% for final testing. The calibration set was kept separate from both model training and final evaluation so that the conformal prediction thresholds could be estimated without using the test data.

3 Baseline CNN and Conformal Prediction

Baseline CNN. Our baseline model is a 3-layer convolutional neural network. The first layer maps the single-channel input to 16 output channels, the second maps those 16 channels to 32, and the final layer maps those 32 channels to 64. Each convolutional layer uses ReLU activation, with max pooling after the first two layers. Global average pooling reduces the final 64-channel representation before binary classification.

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(a) Training and validation loss; lowest validation loss at epoch 62.

(b) Confusion matrix on the 30 held-out candidates.

Figure 1: Baseline CNN results: (a) loss curves, (b) test-set confusion matrix.

We trained the CNN for 250 epochs using a batch size of 32 and used the checkpoint with the lowest validation loss. Epoch 62 gave the lowest validation loss before the model overfitted (Figure 1).

On the 30-candidate test set, the baseline CNN achieved 80% accuracy, 73.68% precision, 93.33% recall, an F1 score of 0.824 (harmonic mean of precision and recall), and a ROC-AUC of 0.844 (how well the model distinguishes the two classes). The resulting confusion matrix is shown in Figure 1.

Conformal prediction layer. We then added a conformal prediction layer to the trained CNN to identify uncertain classifications and defer them for further review. Using a separate calibration set of around 20 examples, we computed the nonconformity score $s = 1 - p_{\text{true}}$, where p_{true} is the probability assigned to the correct class. We then selected a finite-sample-adjusted threshold for each target coverage level of 80%, 90%, and 95%, without retraining the CNN.

For each test example, the prediction set includes every class whose nonconformity score falls at or below the calibration threshold. A prediction set containing exactly one class produces an automatic decision. Sets containing both classes or neither class trigger abstention. We evaluated prediction-set coverage, abstention rate, accuracy among automatic decisions, and false-positive rates on the same 30 held-out examples used for baseline evaluation.

At the 90% target coverage level ($\alpha = 0.10$), the method achieved 90.0% empirical coverage and reduced false positives from five to two while abstaining on 12 of 30 examples. At the 95% target ($\alpha = 0.05$), it achieved 96.7% coverage with zero automatic false positives but abstained on 19 examples. These preliminary results show a tradeoff between reducing false-positive decisions and increasing the number of cases requiring review. Coverage measures whether the prediction set contains the true label; it does not imply a correct automatic classification when both labels are included.

4 Results

Table 1 compares the baseline CNN with conformal prediction at three coverage targets on the clean test set. Table 2 shows how both degrade under added Gaussian noise. At $\alpha = 0.10$, conformal prediction cuts automatic false positives from 5 to 2 at the cost of abstaining on 40% of cases.

5 Interpretations, Limitations, and Next Steps

As synthetic noise increased, CNN accuracy fell from 80% to 72%, and its false-positive rate rose from 33.33% to 52.67%. At $\alpha = 0.10$, conformal coverage fell from 90% to 88%, abstention rose from 40% to 47.33%, and the automatic false-positive rate rose from 13.33% to 24%. Conformal prediction reduced automatic false positives, though it increased deferrals; however, its robustness

Table 1: Clean-test performance. CNN and conformal prediction (CP) on 30 held-out Kepler candidates. Accuracy is measured among automatic decisions, and false-positive rates use all 15 actual negative candidates as the denominator. Stricter coverage targets cut false positives but raise abstention.

Method	Accuracy (decided)	FPR	Abstention rate	Coverage
Baseline CNN	80.00% (24/30)	33.33% (5/15)	0.00% (0/30)	—
CP, $\alpha = 0.20$	80.00% (24/30)	33.33% (5/15)	0.00% (0/30)	80.00%
CP, $\alpha = 0.10$	83.33% (15/18)	13.33% (2/15)	40.00% (12/30)	90.00%
CP, $\alpha = 0.05$	90.91% (10/11)	0.00% (0/15)	63.33% (19/30)	96.67%

Table 2: Synthetic-noise stress test. Original CNN and conformal prediction at $\alpha = 0.10$ under additive Gaussian noise applied after light-curve normalization. The clean row is evaluated once; noisy rows show means over 10 perturbations of the same 30 candidates. Model weights and clean-calibration thresholds were fixed.

Noise level (σ)	CNN acc.	CNN FPR	CP FPR	CP coverage	CP abstention	CP singleton acc.
Clean (0.00)	80.00%	33.33%	13.33%	90.00%	40.00%	83.33%
Low (0.10)	78.67%	36.00%	13.33%	90.00%	40.33%	83.21%
Medium (0.25)	75.33%	42.00%	13.33%	90.33%	46.67%	81.87%
High (0.50)	72.00%	52.67%	24.00%	88.00%	47.33%	77.40%

under distribution shift remains unverified. The main limitations are the small test set of 30 balanced candidates and the use of synthetic rather than realistic noise.

6 Class-Survey Contribution

Case summary. Our project examines the reliability of AI-assisted exoplanet candidate classification. A CNN achieved 80% accuracy on 30 held-out Kepler candidates. At a 90% target coverage level, conformal prediction reduced automatic false positives from five to two while abstaining on 40% of cases. Under high synthetic noise, the CNN's accuracy declined and conformal coverage fell below its target. These findings illustrate the tradeoff between automation and caution, emphasizing the need for realistic stress testing and expert review before operational use.

Comparison-matrix entry. *Domain:* astronomy. *Decision:* exoplanet candidate classification. *AI role:* CNN with conformal abstention. *Requirement:* limit incorrect positive classifications. *Evidence:* held-out KOI evaluation and Gaussian-noise tests. *Boundary:* small, balanced dataset and synthetic perturbations. *Supported conclusion:* fewer automatic false positives at the cost of abstention. *Fallback:* expert review. *Responsible role:* astronomical science-team reviewer. *Open question:* reliability under realistic astronomical distribution shifts.

7 Team Contributions and Remaining Plan

Pranav built the Kepler data pipeline, and created the train, validation, calibration, and test splits used for evaluation. Connor then used those to build the baseline CNN, the model evaluation framework, and related visualizations. Viraj implemented the conformal prediction layer, prediction sets, and the evaluation. Ermuun implemented Gaussian-noise stress testing, examined alternative dataset splits, and produced comparative results.

Following the milestone, we will focus on refining the conformal prediction framework and conducting more extensive experiments to evaluate its effectiveness under different distribution shifts, particularly stellar flares and instrumental noise. We will further analyze the trade-off between classification performance, prediction coverage, and abstention rates, refine our approach based on these findings, and complete the final evaluation and documentation.

Repository. Latest version is in the `data-evaluation-improvements` branch of `https://github.com/HeavyMoonshine/ECE_4424_Fall_2026_Project`.

References

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[Lightkurve developers, 2026] [NASA Exoplanet Science Institute, 2018]